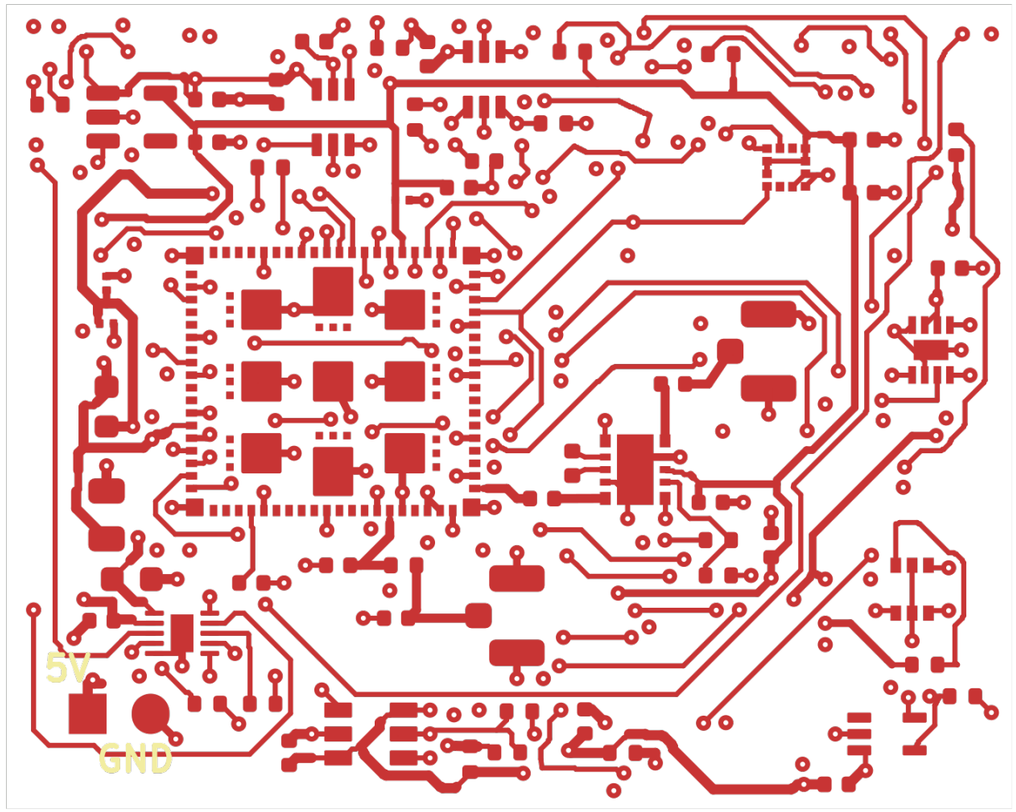
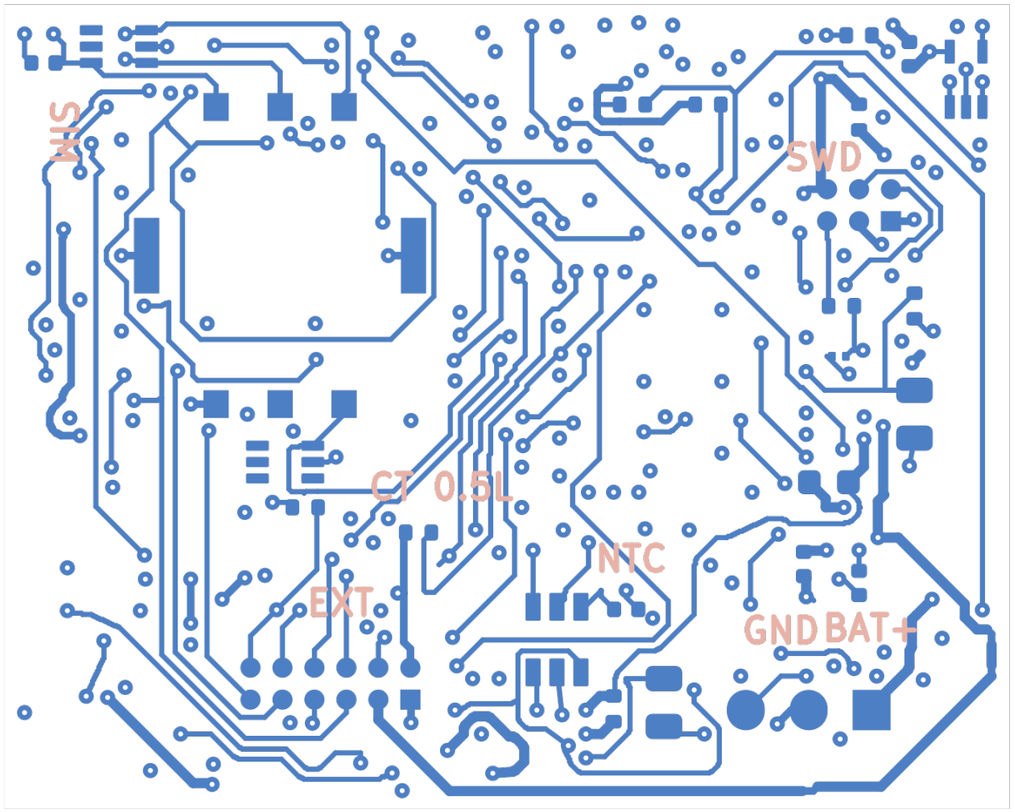
COLLAR TRACKER / r05l PCB REVIEW

Charge interlock implemented. Two clean PCB DFM reports; assembled release remains on hold.

TOP / native copper and silkscreen



BOTTOM / mirrored component view



VERIFIED PCB

40 × 32 mm / six Cu layers / 87 footprints

Native DRC, ERC, connectivity and parity: zero findings

Two JLC runs: 0 Danger / 0 Warning / 455 Good each

Same ZIP, same rules, all 29 rows analyzed; SMT off

81 fitted references / 44 exact candidate MPNs

109 physical GND pads have local paths to vias

Two GND planes; 154/154 signal-via return screens pass

REQUIRED BEFORE WORKING DELIVERY

Charge window is implemented; corner model is conditional.

Thermal coupling, startup and fault transients need tests.

Protected pack, NTC attachment and antennas: open.

Firmware and factory programming/functional test: open.

Procurement, CPL rotations and SMT acceptance: open.

No physical powered, RF or water/dust result exists.

This package is for review; it is not an order release.

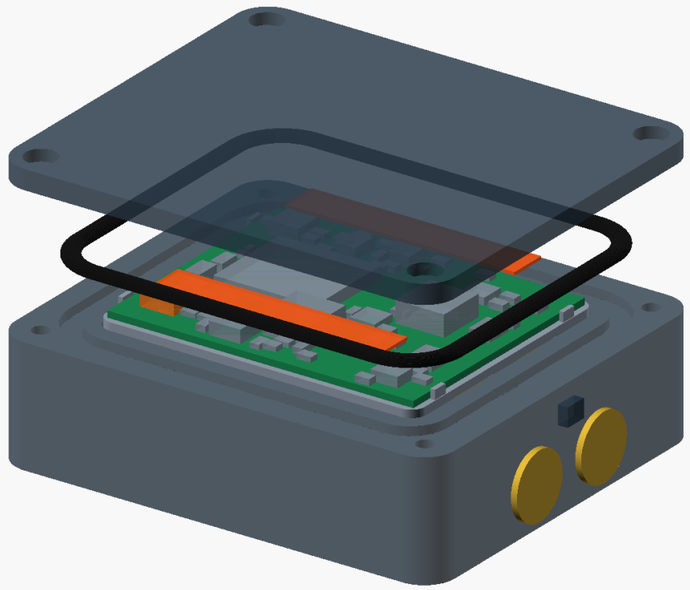
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JLC records: 637366592129687553 / 637366801595977730

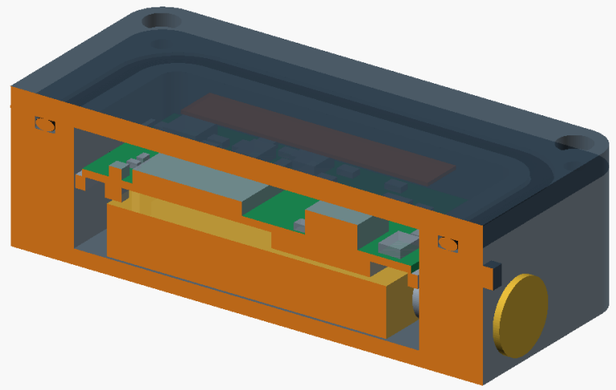
COLLAR TRACKER / r07 ENCLOSURE REVIEW

Actual PCB placements and carrier clearance. Pack and antenna boxes remain reserved volumes.

EXPLODED / shell, seal and populated envelopes



SECTION / internal clearance model



VERIFIED CAD GEOMETRY

85 populated envelopes; four PCB thickness cases pass.

New U12 relief and one moved support leg; 1 mm bridge.

Actual CAD clearance intersection is empty.

Previous carrier control has 1.234 mm³ interference.

Eight exported parts: closed, oriented and connected.

Rigid body + lid: 57.4 × 49.4 × 17.6 mm (key excluded).

Boot envelope: 60.4 × 52.4 × 22.4 mm.

WATER / DUST TARGET — UNQUALIFIED

IP68 swimming/dust target; no physical test yet.

Provisional immersion: 2 m fresh water / 60 minutes.

Pack and antenna dimensions still need real selections.

Material/seal grades and fastener torque remain open.

Harness, dock wiring and positive PCB retention: open.

Carrier strength and thermal coupling require tests.

Grey/orange blocks are fit bounds, not qualified parts.

Model: mechanical/swim-r07-charge-interlock; exact input hashes in mesh-checks.json.

PCB and enclosure evidence are separate from assembled electrical and ingress acceptance.